

لا صفا مزه
الكمه
المعتمدين على الهم
يكون
الاشين و
السجاد

Al-Balqa' Applied University



جامعة البلقاء التطبيقية

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Date: Nov. 27, 2017

Electrical circuits 1.

Mid Term
Exam First semester
2017-2018

Exam Time: 2 hours

Choose the correct answer.

1) A circuit element draws 15 A when connected to a 240-V line. How long does it take to consume 180 kJ?

- a) 5 ms b) 0.5 ms c) 2.67s

d) 50 s

2) To move charge $q = 6$ C from point a to point b requires -30 J. The voltage drop V_{ab} is:

- a) 5V b) 180 V

c) -5 V

d) -180V

3) The total charge entering a terminal is given by $q = (10 - 10e^{-2t})$ mC. The current at $t = 1$ s is:

- a) 1.353 A

b) 2.707A

c) -0.73A

d) 0.73A

4) For the circuit in fig. 1 the voltage v_1 is:

- a) 8V

b) 32V

c) 16V

d) -8V

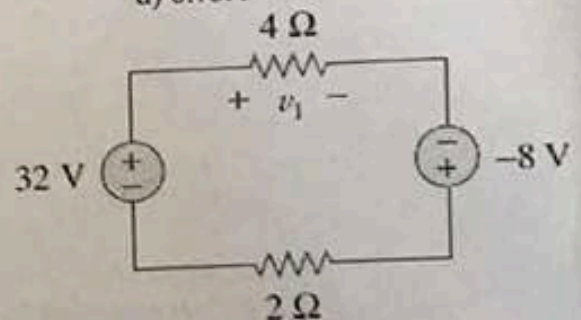


Fig. 1

* For the questions 5,6,7 consider the circuit shown in figure 2 and answer:

5) The equivalent resistance R_{eq} between the terminals a-b of the current source is:

- a) 6Ω b) 6.67Ω

c) 20Ω

d) 12 Ω

6) The current I_0 in the 6-Ω resistor is:

- a) 2A

b) 1A

c) 1.5A

d) 3A

7) The voltage across 8Ω resistor is:

- a) 72V

b) 60 V

c) 240V

d) 120V

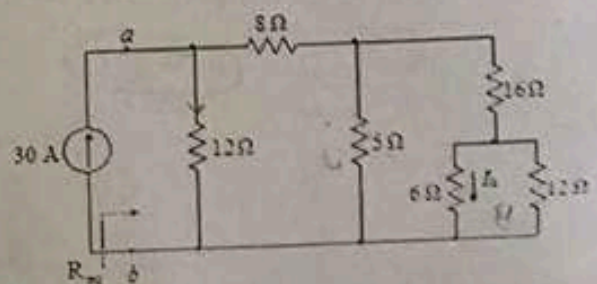


Fig.2

* For the questions 8,9,10,11,12 consider the circuit shown in figure 3 and answer:

8) The value of V_1 is:

a) 36.5V

c) 365 mV

d) -4.8V

b) -18.125V

9) The value of V_2 is:

a) 5V

c) 2.5V d) -500mV

b) 200mV

10) The value of the current (i) in 2Ω resistor is:

a) 0.74A

c) -0.7A

d) 3.625A

b) 1.48A

11) The power absorbed by 4Ω resistor is:

a) 1.563W

c) 5.23W

b) 10.46W

d) 6.46W

12) The power delivered by 2 A current source is:

a) 4W

b) 5W

c) 25W d) -44.8W

* For the questions 13,14,15 consider the circuit shown in figure 4 and answer:

13) The value of V_x is:

a) 5V

c) 20V

b) 10V

d) 100V

14) The current in 20Ω resistor is:

a) 1A

b) 200 mA

c) 2A d) 2.8A

15) The delivered power by the 4 A current source is:

a) 6W

c) 20 W

b) 16 W

d) 40W

16) Refer to figure 5; the value of the equivalent resistance is:

a) 62.5 Ω

c) 45.5 Ω

b) 36.25 Ω

d) 75 Ω

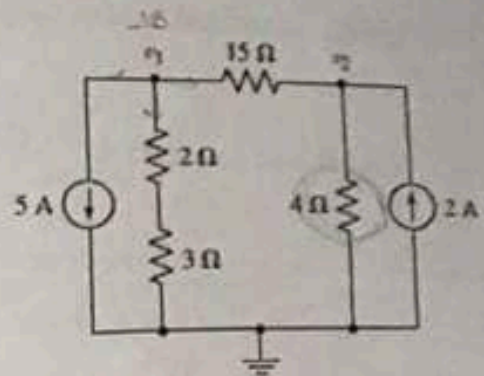


Fig.3

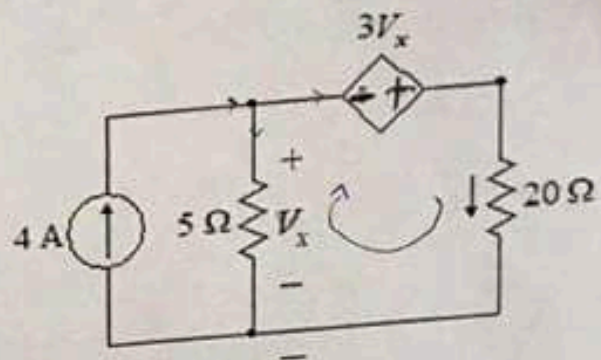


Fig.4

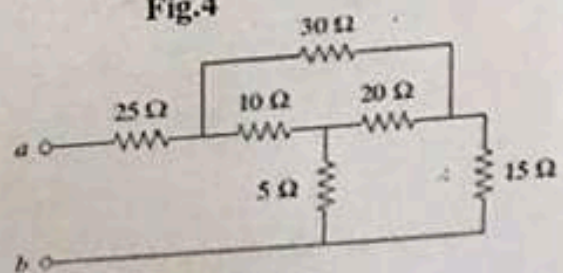
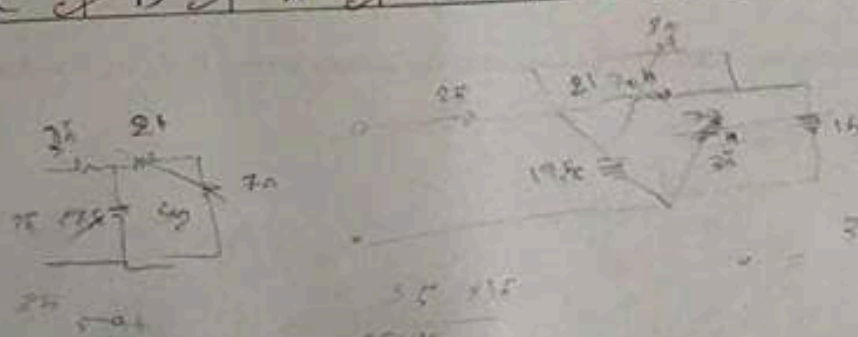


Fig 5

1	2	3	4	5	6	7	8
A	A	B	C	C	A	D	B
9	10	11	12	13	14	15	16
C	D	A	B	B	C	D	B

Good Luck



①

$$1) I = 15A \quad V = 240V \quad W = 180 kJ \quad t = ??$$

sol

$$P = \frac{dW}{dt} \rightarrow IV = \frac{dW}{dt} \rightarrow t = \frac{180 \times 10^3}{240 \times 15} = \boxed{50 s}$$

$$2) q = 6C \text{ move from a to b} \quad W = -30J \quad V_{ab} = ??$$

sol

$$W = -30J$$

الأشارة السالبة تعني أن الشحنة الموجبة انتقلت من نقطة ذات جهد مرتفع إلى نقطة ذات جهد منخفض

$$\therefore V_a > V_b$$

$$V_{ab} = V_a - V_b \rightarrow \therefore V_{ab} = \oplus \quad V_{ab} = \frac{dW}{dq} = \frac{30}{6} = \boxed{5V}$$

$$3) q = (10 - 10e^{-2t}) mC \quad \text{find } I \text{ at } t = 1s$$

sol

$$I = \frac{dq}{dt} = \frac{d(10 - 10e^{-2t}) m}{dt} = -2 \times -10e^{-2t} m$$

$$\therefore I = 20e^{-2t} m \quad \text{at } t = 1s \rightarrow$$

$$I = 20e^{-2} m = \boxed{2.707 mA}$$



⑦

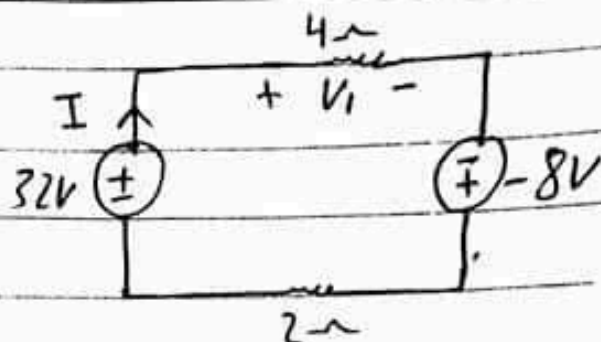
4) find $V_1 \rightarrow$

soln

$$-32 + 4I - (-8) + 2I = 0$$

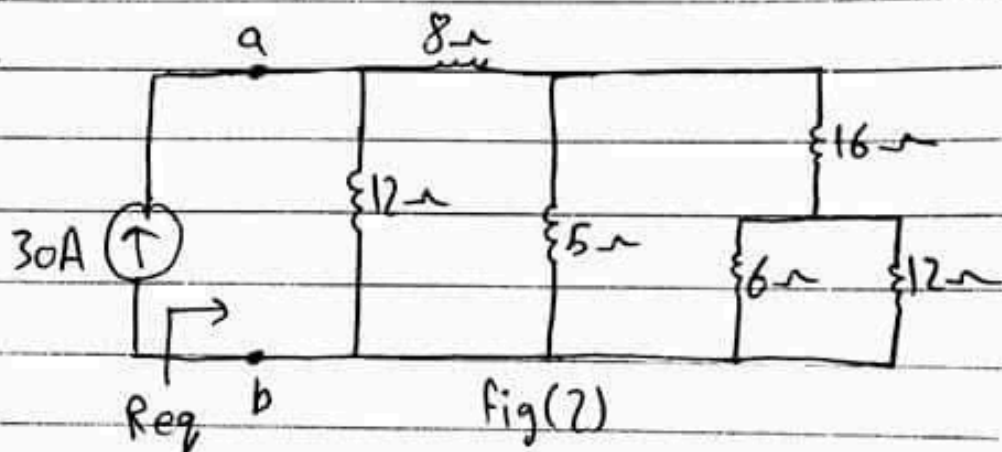
$$6I = 24 \rightarrow I = 4A$$

$$V_1 = IR = 4 \times 4 = \boxed{16V}$$



5) find $R_{eq} \rightarrow$

soln



- 1) $12 // 6 = 4\Omega$
- 2) $(4 \text{ with } 16) = 20\Omega$
- 3) $20 // 5 = 4\Omega$
- 4) $(4 \text{ with } 8) = 12\Omega$
- 5) $12 // 12 = \boxed{6\Omega}$



③

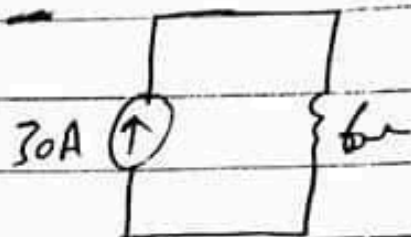
6) For fig(2) ↓ I in 6Ω →

solⁿ

find

* هنا نخرج بالدارة، وجوب لمعرفة تيار، وجهد كل مقاومة

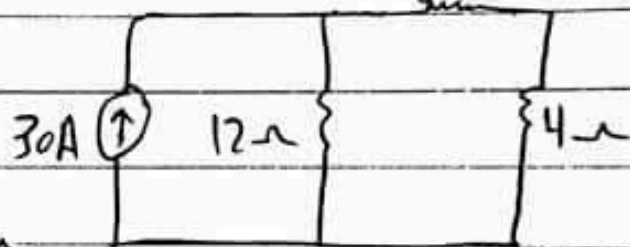
$$V_{6\Omega} = 30 \times 6 = 180 = V_{12\Omega} = V_{12\Omega}$$



$$V_{12\Omega} = IR \rightarrow I = \frac{180}{12} = 15A$$

8Ω

$$V_{8\Omega} = 120V \quad V_{4\Omega} = 60V = V_{20\Omega} = V_{5\Omega}$$



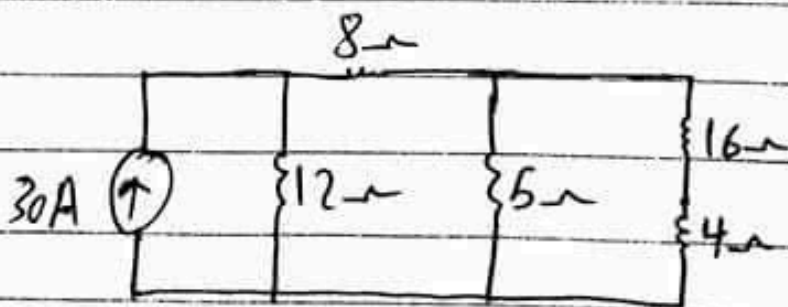
$$V_{20\Omega} = 60V = IR$$

$$\therefore I = \frac{60}{20} = 3A$$

$$V_{4\Omega} = 3 \times 4 = 12V$$

$$= V_{12\Omega} = V_{6\Omega}$$

$$I_{6\Omega} = \frac{V_{6\Omega}}{R} = \frac{12}{6} = 2A$$

7) for fig(2) find $V_{8\Omega}$ →solⁿ

$$V_{8\Omega} = 120V$$

الجواب في الفرع السابق

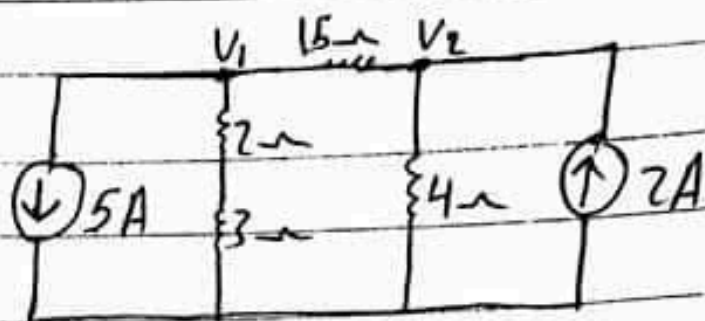


④

for next figure find →

8) find V_1 → 9) V_2 →
soln

Using node theorem →



at node 1 →

$$\frac{V_1 - V_2}{15} + \frac{V_1}{2+3} + 5 = 0 \quad \text{--- (1)}$$

at node 2 →

$$\frac{V_2 - V_1}{15} + \frac{V_2}{4} = 2 \quad \text{--- (2)}$$

$$\therefore \boxed{V_1 = -18.125 \text{ V}}$$

$$\boxed{V_2 = 2.5 \text{ V}}$$

10) find the current in 2Ω →soln calculate V_1 at

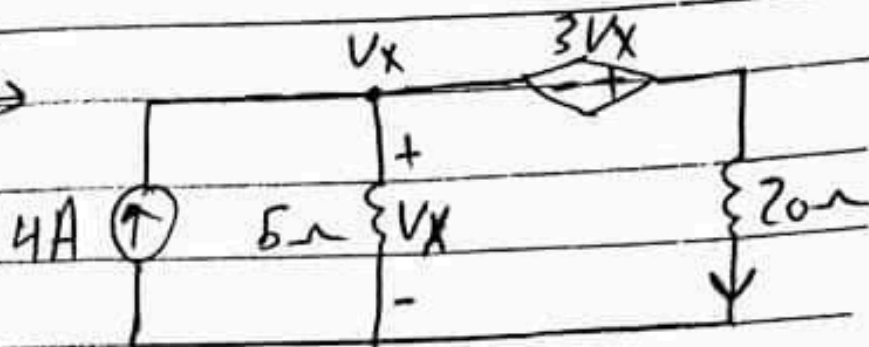
$$I_{2\Omega} = \frac{0 - V_1}{R} = \frac{0 - (-18.125)}{2+3} = \boxed{3.625 \text{ A}}$$

$$11) \text{ Power in } 4\Omega \rightarrow P = \frac{V^2}{R} = \frac{(2.5)^2}{4} = \boxed{1.5625 \text{ W}}$$

$$12) \text{ power in (2A source)} \rightarrow P = IV = I \times V_2 \\ = 2 \times 2.5 = \boxed{5 \text{ W}}$$



⑤

for next figure find $\rightarrow$ 13) V_x 14) $I_{20\Omega}$ 

Soln

Using Node theorem $\rightarrow$

$$\frac{V_x}{5} + \frac{V_x + 3V_x}{20} = 4 \rightarrow \frac{V_x}{5} + \frac{V_x}{5} = 4 \rightarrow \boxed{V_x = 10V}$$

$$I_{20\Omega} = \frac{V_x + 3V_x}{20} = \frac{10 + 3 \times 10}{20} = \frac{40}{20} = \boxed{2A}$$

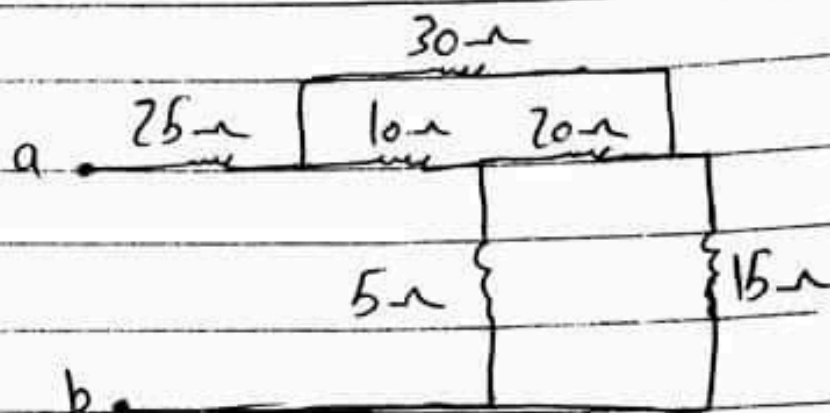
15) power in (4A source) $\rightarrow$

$$P = IV = I \times V_x = 4 \times 10 = \boxed{40W}$$



⑥

16) find $R_{eq} \rightarrow$



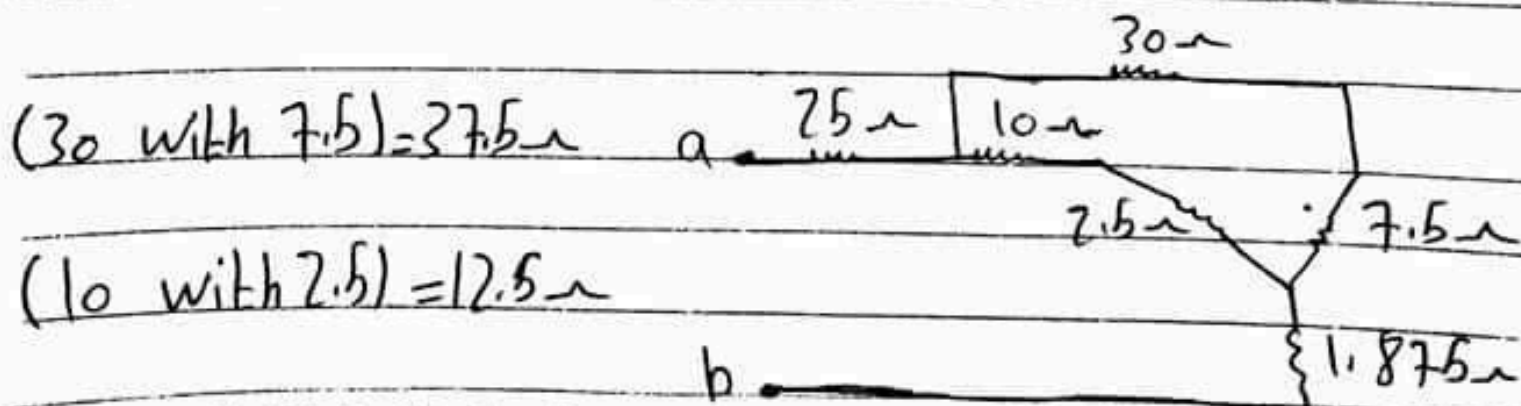
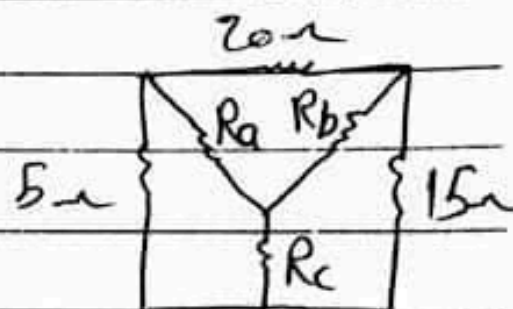
Solⁿ

Using $\Delta - Y \rightarrow$

$$R_a = \frac{20 \times 5}{5 + 20 + 15} = 2.5 \Omega$$

$$R_b = \frac{20 \times 15}{5 + 20 + 15} = 7.5 \Omega$$

$$R_c = \frac{15 \times 5}{5 + 20 + 15} = 1.875 \Omega$$



$$(30 \text{ with } 7.5) = 37.5 \Omega$$

$$(10 \text{ with } 2.5) = 12.5 \Omega$$

$$37.5 // 12.5 = 9.375 \Omega$$

$$(25 \text{ with } 9.375 \text{ with } 1.875) = 36.25 \Omega$$





Choose the correct answer.

1) A light bulb changes its resistance as it heats. What is the resistance of a 120 V, 60 W bulb when it has been on for a few minutes:

- a) 240 Ω ✓
b) 2400 Ω ✓
c) 5 Ω ✓
d) 0.5 Ω ✓

2) If a conductor with a current of 175mA converts 45 J of electrical energy into heat in 25 sec, what is the potential drop across the conductor:

- a) 9.3 V
b) 10.3 V ✓
c) 11.3 V
d) 12.3 V

3) Find the conductance of a 1.5 M Ω resistor:

- a) 66.7 mS
b) 6.67 mS
c) 667 mS
d) 0.667 mS ✓

* For the questions 4,5,6 consider the circuit shown in figure 1 and answer:

4) The equivalent resistance R_{eq} between the terminals a-b of the current source is:

- a) 21.36 Ω ✓
b) 8.54 Ω
c) 31.46 Ω
d) 25.2 Ω

5) The current I_0 in the 20- Ω resistor is:

- a) 1.55 A
b) 1.42 A
c) 4.55 A ✓
d) 3.5 A

6) The voltage across 30 Ω resistor is:

- a) 40.9 V
b) 81.8 V
c) 63.5 V
d) 163.5 V ✓

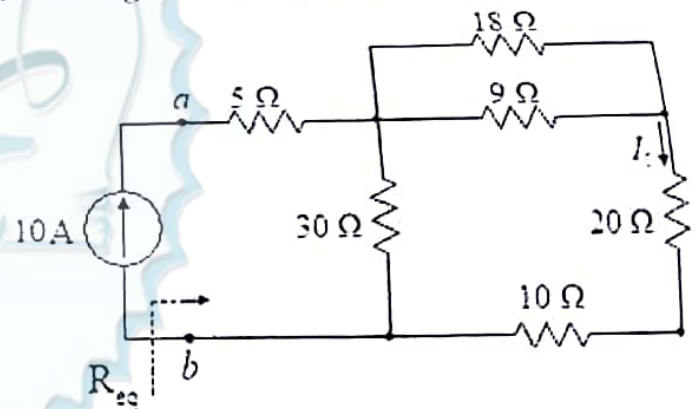


Fig.1

* For the questions 7,8,9,10,11 consider the circuit shown in figure 2 and answer:

7) The value of V_1 is:

- a) 3.5V
b) 4.5V
c) 365 mV
d) 7.5V

8) The value of V_2 is:

- a) 2.5V
b) -500mV
c) 1.5V
d) 68.5mV

9) The value of the current (ia) in $6k\Omega$ resistor is:

- a) 0.75A
b) 0.35A
c) 0.5A
d) 0.25mA

10) The power absorbed by $6V$ voltage source is:

- a) -3.75mW
b) 562.5mW
c) 52.33 mW
d) -5.573mW

11) The power delivered by the dependent voltage source is:

- a) 45mW
b) 0.375mW
c) -0.255 mW
d) -44.8mW

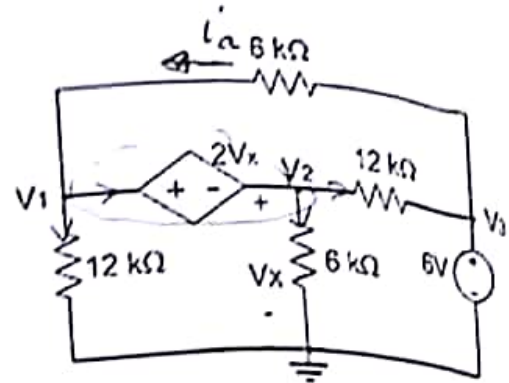


Fig.2

* For the questions 12,13,14 consider the circuit shown in figure 3 and answer:

12) The value of V_b is:

- a) 6V
b) 4.5V
c) -6V
d) 24V

13) The current in 8Ω resistor is:

- a) 1.5A
b) 200 mA

- c) 3A
d) 6A

14) The delivered power by the 21 V voltage source is:

- a) -6 W
b) 36 W
c) 24 W
d) 63 W

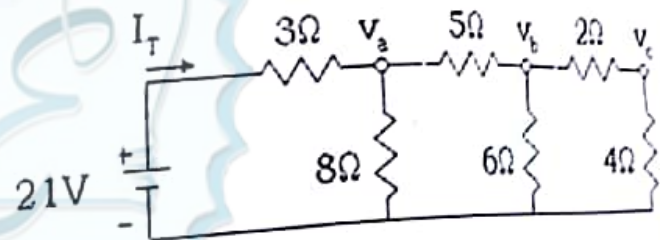


Fig.3

15) Refer to figure 4; the value of the equivalent resistance is:

- a) 11.85 Ω
b) 17.5 Ω
c) 16 Ω
d) Ω

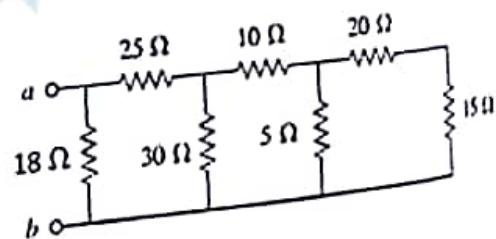


Fig 4

16) How many joules of energy will a 10Ω resistor connected across a 15V battery dissipate in 1 minute?

- a) 13.5kJ
b) 135J
c) 0.135kJ
d) 135 J

17) The power 2,000,000,000 W can be expressed in powers of 10 as:

- a) 2 MW (b) 2 TW (c) 2 GW d) 2 PW

18) A charge of 2 C flowing past a given point each microsecond is a current of:

- A) 2k A b) 200 mA
c) 2 mA (d) 2 MA

19) The resistance of a conductor of diameter d and length l is $R \Omega$. If the diameter of the conductor is halved and its length is doubled, the resistance will be:

- a) $1R \Omega$ b) $2R \Omega$
(c) $4R \Omega$ d) $8R \Omega$

20) How many coulombs of charge flow through a circuit carrying a current of 10 A in 1 minute:

- a) 10 C b) 60 C
(c) 600 C d) 1200 C

1	2	3	4	5	6	7	8	9	10
a	b	d	a	c	d	a	a	c	a
11	12	13	14	15	16	17	18	19	20
c	b	a	d	a	b	b	d	c	c

Good Luck



Choose the correct answer.

** Refer to figure 1; and answer the questions 1,2,3,4

1) the voltage V_1 is:

- a) 60V b) 50V
c) 15V d) 45V

2) the voltage V_2 is:

- a) 60V b) 15V
c) 50V d) 75V

3) the power dissipated in $3k\Omega$ is:

- a) 675mW b) 17.5W
c) 160mW d) 607 kW

4) the power supplied by the current source is:

- a) 12W b) 175 mW
c) 160mW d) 1.8 W

5) A heater element draws 15 A when connected to a 240-V line. How long does it take to consume 180 kJ?

- a) 5 ms b) 0.5 s c) 50 ms d) 50 s

6) The total charge entering a terminal between $t=1s$ and $t=2s$ if the current passing the terminal is $i(t)=3t^2$ A is:

- a) 7C b) 90 C c) -5 C d) 90 mC

7) The power 2,000,000 W can be expressed in powers of 10 as:

- a) 2 MW b) 2 GW c) 2 TW d) 2 PW

8) A charge of 200mC flowing past a given point each millisecond is a current of:

- A) 2 A b) 200 A
c) 2 mA d) 2 kA

9) The maximum current that a 2W, 80 kΩ resistor can safely conduct is

- a) 160 kA b) 40 kA
c) 5 mA d) 125 mA

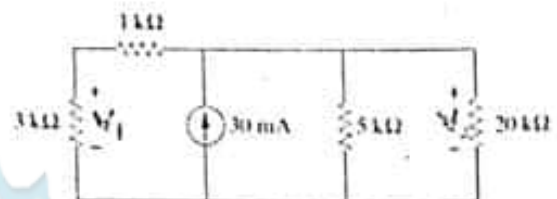


Fig. 1

* For the questions 10,11,12 consider the circuit shown in figure 2 and answer:

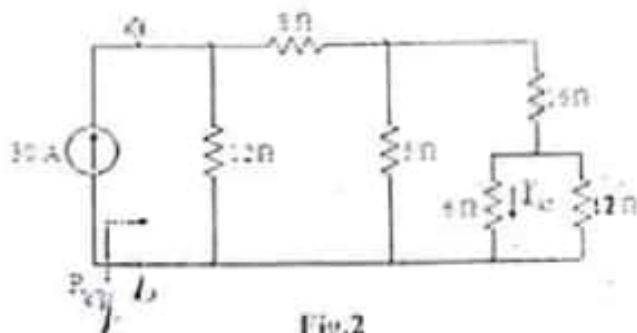


Fig.2

10) The equivalent resistance R_{eq} between the terminals a-b of the current source is:

- a) 6Ω b) 8Ω
c) 10Ω d) 4Ω

11) The current I_0 in the 6Ω resistor is:

- a) $4A$ b) $1.5A$
c) $2A$ d) $3A$

12) The voltage across 8Ω resistor is:

- a) $140V$ b) $120V$
c) $60V$ d) $150V$

* For the questions 13,14,15,16,17 consider the circuit shown in figure 3 and answer (use nodal analysis method):

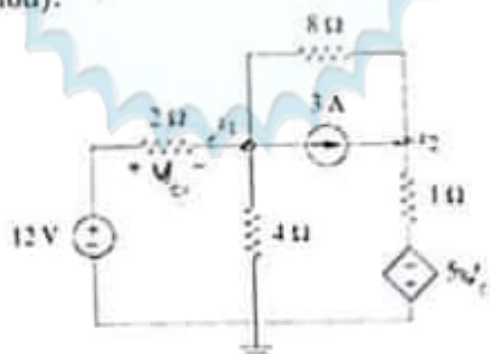


Fig.3

13) The value of V_1 is: ~~3A~~ ~~37V~~ 37V

14) The value of V_2 is: 36V

15) The value of the current (i_o) in $2\ \Omega$ resistor is: 6A

16) The power absorbed by $4\ \Omega$ resistor is: $18\ W$

17) Write down the nodal equations (don't simplify):

~~$V_1 = 12V$~~
 ~~$V_1 - 12 = \frac{V_1}{4} + \frac{V_1 - V_2}{8}$~~

$\sum V = 0 \quad \frac{V_1 - 12}{2} + \frac{V_1}{4} + \frac{V_1 - V_2}{8} = 0$

* For the questions 18,19,20 consider the circuit shown in figure 4 and answer (use mesh analysis method):



18) The current i is: 3A

19) The current i is: 5A

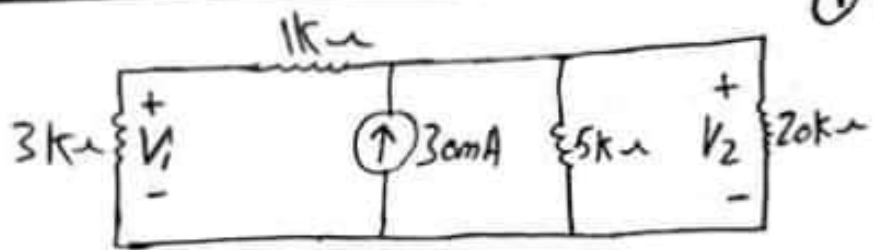
20) The delivered power by the 12 V voltage source is: -36W

1	2	3	4	5	6	7	8	9	10
d	a	a	d	d	a	a	b	c	a
11	12	13	14	15	16	17	18	19	20
C	b	37V	36V	6A	$18\ \Omega$	37V	1.3A	5A	-36W

Good Luck

Find $\rightarrow V_1, V_2$
 Power for $3k\Omega$
 Power for current source

1, 2, 3, 4



①

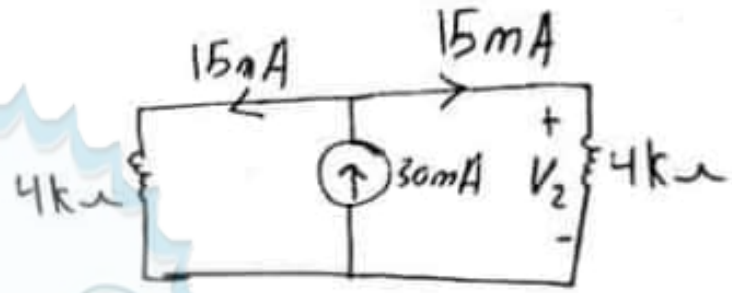
soln

$$20k \parallel 5k = 4k\Omega$$

$1k\Omega, 3k\Omega$ in series $\rightarrow 4k\Omega$

← يتوزع التيار بالتساوي

$$V_2 = IR = 15m \times 4k = 60V$$



$$I_{3k\Omega} = 15mA \quad (\text{توازي})$$

$$\therefore V_1 = IR = 15m \times 3k = 45V$$

$$P_{3k\Omega} = IV = 15m \times 45 = 675mW$$

$$\text{Power for current source} = IV = 30m \times 60 = 1800mW$$

60V ← 4kΩ ← جهد مصدر التيار نفس جهد المقاومة
 لأنهم على التوازي

5) $I = 15A$ $V = 240V$ $W = 180kJ$ $t = ???$

soln

$$P = IV = 15 \times 240 = 3600W$$

$$P = \frac{dW}{dt} \rightarrow dt = \frac{dW}{P} \rightarrow \therefore t = \frac{180kJ}{3600} = 50s$$

(2)

6) $i(t) = 3t^2 \text{ A}$ $t = 1 \text{ s} \rightarrow 2 \text{ s}$ $q = ??$

soln

$$q = \int_{t_0}^t i \, dt \rightarrow q = \int_1^2 3t^2 \, dt \rightarrow q = [t^3]_1^2$$

$$\therefore \boxed{q = 7 \text{ C}}$$

7) $2000,000 \text{ W} \rightarrow \boxed{2 \text{ MW}}$

8) $q = 200 \text{ mC}$ $t = 1 \text{ ms}$ $i = ??$

soln

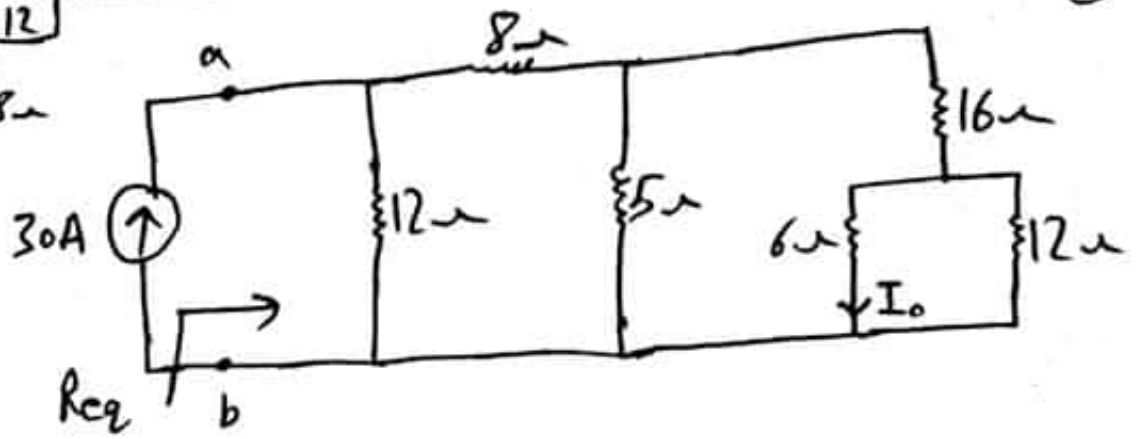
$$i = \frac{dq}{dt} = \frac{200 \text{ m}}{1 \text{ m}} = \boxed{200 \text{ A}}$$

9) $P = 2 \text{ W}$ $R = 80 \text{ k}\Omega$ $i = ??$

soln

$$P = i^2 R \rightarrow i^2 = \frac{P}{R} \rightarrow i^2 = \frac{2}{80 \text{ k}} \rightarrow \boxed{i = 5 \text{ mA}}$$

Find R_{eq} , I_0 , $V_{8\Omega}$



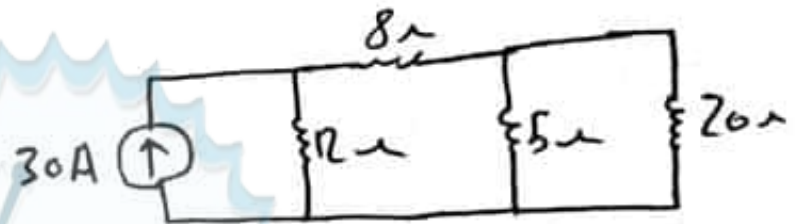
soln

$$12 // 6 = 4\Omega \rightarrow 4, 16 \text{ in series} \rightarrow 20\Omega$$

$$20 // 5 = 4\Omega$$

$$4, 8 \text{ in series} \rightarrow 12\Omega$$

$$12 // 12 = 6\Omega = R_{eq}$$



* كل مقاومة بحسب فيها تيار $15A$ لأنهم متساوية

~~$$I_{12\Omega} = I_{8\Omega} = I_{4\Omega} = 15A$$~~

$$I_{12\Omega} = I_{8\Omega} = I_{4\Omega} = 15A \quad (\text{توازي})$$

$$V_{8\Omega} = IR = 8 \times 15 = 120V$$

$$V_{4\Omega} = IR = 15 \times 4 = 60V = V_{20\Omega} = V_{5\Omega}$$

$$I_{20\Omega} = \frac{V}{R} = \frac{60}{20} = 3A = I_{16\Omega} = I_{4\Omega}$$

$$V_{4\Omega} = IR = 3 \times 4 = 12V = V_{6\Omega} = V_{12\Omega}$$

$$I_0 = I_{6\Omega} = \frac{V}{R} = \frac{12}{6} = 2A$$

13-17

(4)

soln

at node 1 $\rightarrow$

$$\frac{V_1 - 12}{2} + \frac{V_1}{4} + \frac{V_1 - V_2}{8} + 3 = 0$$

$$0.875V_1 - 0.125V_2 = 3 \quad \text{--- (1)}$$

at node 2 $\rightarrow$

$$\frac{V_2 - V_1}{8} + \frac{V_2 + 5V_0}{1} = 3$$

$$-0.125V_1 + 1.125V_2 + 5V_0 = 3 \quad \text{--- (2)}$$

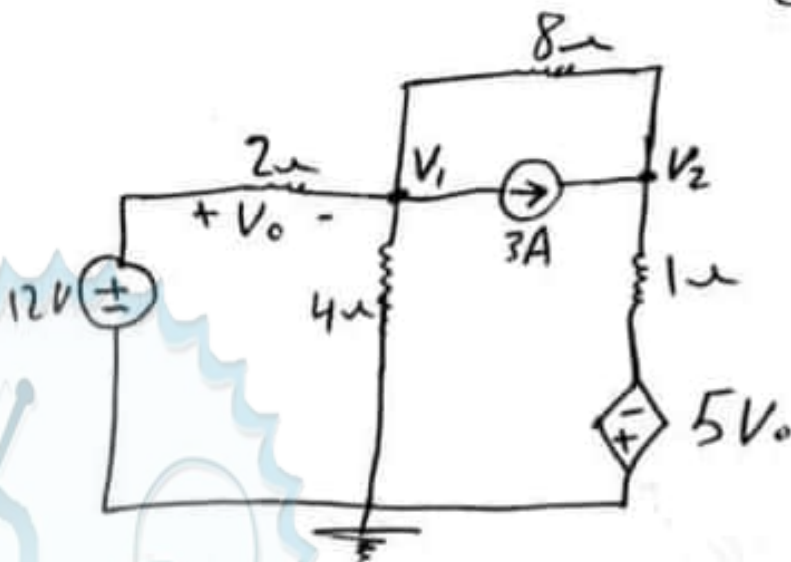
$$V_0 = 12 - V_1 \quad \rightarrow \quad V_1 + V_0 = 12 \quad \text{--- (3)}$$

* هذه الآلة الحاسبة $\leftarrow$

$$V_1 = -10.91 \text{ V}$$

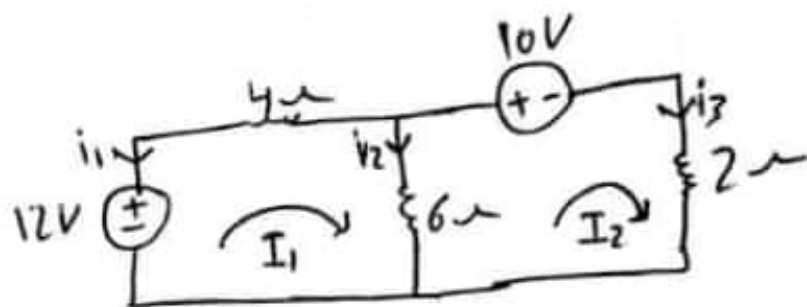
$$V_2 = -100.36 \text{ V}$$

$$V_0 = 22.91 \text{ V}$$



18-20

Find i_1, i_3 , Power
for Voltage source $\rightarrow$



Soln

~~at~~

at loop 1 $\rightarrow$

$$-12 + 4I_1 + 6(I_1 - I_2) = 0 \rightarrow 10I_1 - 6I_2 = 12 \quad (1)$$

at loop 2 $\rightarrow$

$$6(I_2 - I_1) + 10 + 2I_2 = 0 \rightarrow -6I_1 + 8I_2 = -10 \quad (2)$$

$$\therefore I_1 = 0.818 \text{ A} \quad I_2 = -0.636 \text{ A}$$

$$i_1 = -I_1 = \boxed{-0.818 \text{ A}} \quad i_3 = I_2 = \boxed{-0.636 \text{ A}}$$

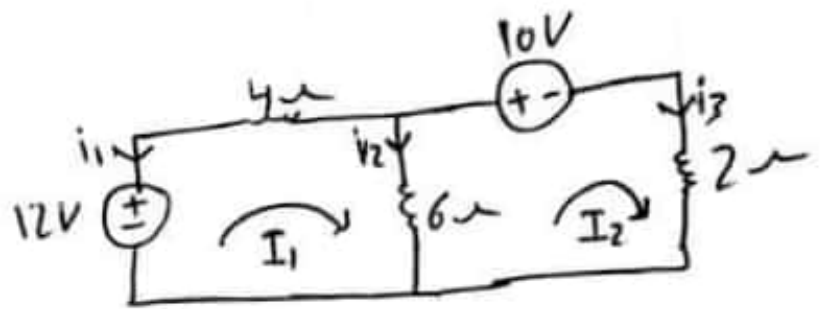
Power in Voltage source $\rightarrow$

$$P = iV = 0.818 \times 12 = \boxed{9.816 \text{ W}}$$

- 1) D 2) A 3) A 4) D 5) D 6) A 7) A
 8) B 9) C 10) A 11) C 12) B 13) -10.91 V
 14) -100.36 V 15) 11.455 A 16) 29.76 W
 17) الجهد 18) -0.818 A 19) -0.636 A 20) 9.816 W

18-20

Find i_1, i_3 , Power
for Voltage source $\rightarrow$



Soln

~~at~~

at loop 1 $\rightarrow$

$$-12 + 4I_1 + 6(I_1 - I_2) = 0 \rightarrow 10I_1 - 6I_2 = 12 \quad (1)$$

at loop 2 $\rightarrow$

$$6(I_2 - I_1) + 10 + 2I_2 = 0 \rightarrow -6I_1 + 8I_2 = -10 \quad (2)$$

$$\therefore I_1 = 0.818 \text{ A} \quad I_2 = -0.636 \text{ A}$$

$$i_1 = -I_1 = \boxed{-0.818 \text{ A}} \quad i_3 = I_2 = \boxed{-0.636 \text{ A}}$$

Power in Voltage source $\rightarrow$

$$P = iV = 0.818 \times 12 = \boxed{9.816 \text{ W}}$$

- 1) D 2) A 3) A 4) D 5) D 6) A 7) A
 8) B 9) C 10) A 11) C 12) B 13) -10.91 V
 14) -100.36 V 15) 11.455 A 16) 29.76 W
 17) الجهد 18) -0.818 A 19) -0.636 A 20) 9.816 W



Choose the correct answer.

• For the questions 1,2,3 consider the circuit shown in figure 1 and answer:

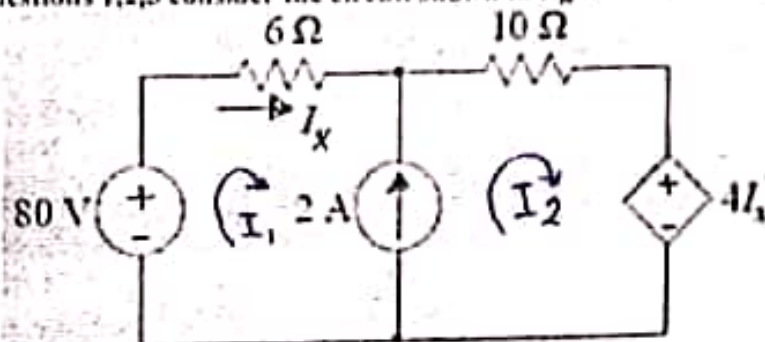


Fig.1

-Write down the mesh equations (don't simplify):

$$\begin{aligned} I_2 - I_1 &= 2 \quad (1) \\ -80 + 6I_1 + 10I_2 + 4I_x &= 0 \quad (2) \end{aligned} \quad \boxed{I_x = I_1}$$

1) The value of I_x is:

- a) -6A b) 2.4 V
c) 3A d) 12 V

2) The current in 10 Ω resistor is:

- a) 200 mA b) 5A
c) 2A d) 4A

3) The delivered power by the 80 V voltage source is:

- a) -72 W b) 360 W
c) 180 W d) 240W

4) Refer to figure 2; the value of the equivalent resistance at terminals ab is:

- a) 130 Ω b) 175Ω
c) 160 Ω d) 260 Ω

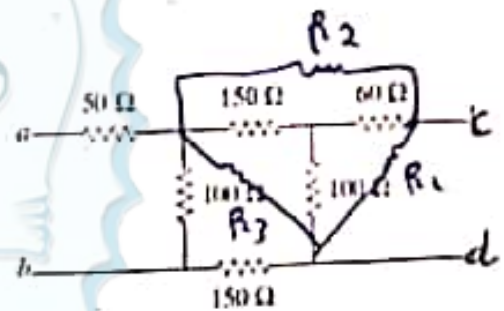


Fig 2

5) Refer to figure 2; the value of the equivalent resistance at terminals cd is:

- a) 160 Ω b) 175 Ω
☒ c) 140 Ω d) 260 Ω

6) A circuit element draws 30 A when connected to a 200-V line. How much energy does it consume for 5 s?

- a) 3kJ b) 3000J c) 6kJ ☒ d) 10kJ

7) The total charge entering a terminal is given by $q = (10 - 10e^{-3t})$ mC.

Calculate the current at $t = 1.0$ s.

- a) 0.5 A b) 90 mA ☒ c) 2.707 mA d) -2.9A

8) The energy 10,000,000,000 Wh can be expressed in powers of 10 as:

- a) 10 MWh ☒ b) 10 GWh c) 10 TWh d) 10 PWh

* For the questions 9,10,11 consider the circuit shown in figure 3 and answer:

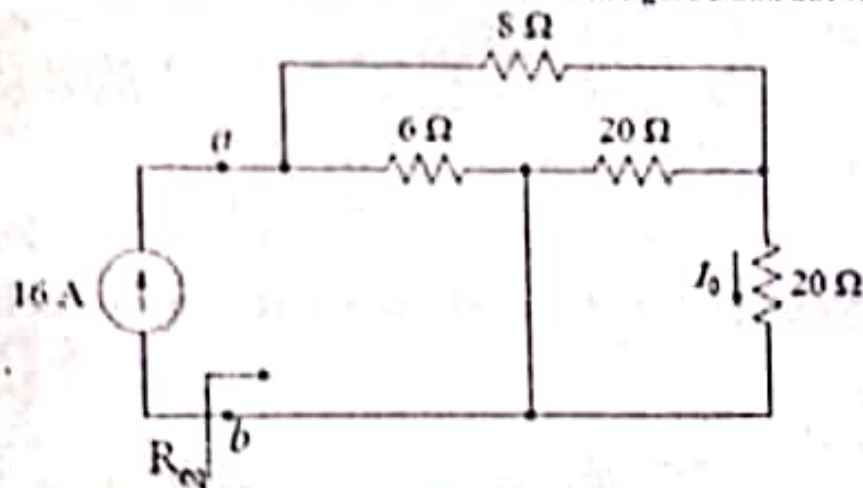


Fig.3

9) The equivalent resistance R_{eq} between the terminals a-b of the ^{current} source is:

- a) 3.4 Ω ☒ b) 4.5 Ω
c) 6 Ω d) 23.4 Ω

10) The current I_0 in the 20- Ω resistor is:

- ☒ a) 2A b) 1.42A
c) 0.5A d) 3A

11) The voltage across the 8 Ω resistor is:

- a) 8 V b) 16 V
c) 24V ☒ d) 32 V

* For the questions 12,13,14,15,16 consider the circuit shown in figure 4 and answer:

-Write down the nodal equations (don't simplify):

$$\frac{V_1 - V_2}{6} + \frac{V_1}{2} = 2 \quad (1)$$

$$\frac{V_2 - V_1}{6} + 3i_o + 2 = \frac{V_2 - 24}{8} = 0$$

$$i_o = \frac{V_1 - V_2}{6} \quad (3)$$

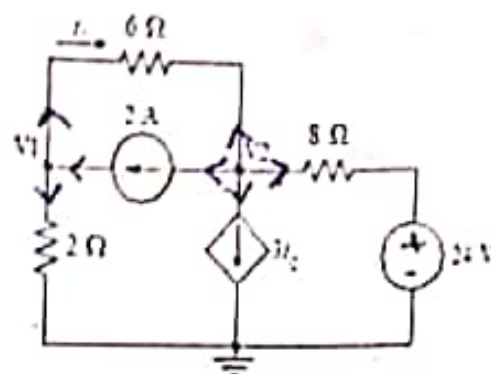


Fig.4

12) The value of V_1 is:

- a) 500 mV b) 1.5 V
☒ c) 3 V d) -7.5V

13) The value of V_2 is:

- ☒ a) 0V b) 200mV
 c) 0.5 V d) 1.5V

14) The value of the current (i_o) in 6 Ω resistor is:

- a) -0.7A ☒ b) 0.5A
 c) 1.5A d) 3 A

15) The power absorbed by 2 Ω resistor is:

- a) 5200 mW ☒ b) 4.5W
 c) 25W d) -55W

16) The power delivered by 24 V voltage source is:

- a) -44.8W b) -36 W c) 36 W ☒ d) 72W

1	2	3	4	5	6	7	8
c	b	d	a	c	d	c	b
9	10	11	12	13	14	15	16
b	a	d	c	a	b	b	d

Good Luck



أسئلة سنوات سابقة

المادة: سيركت ١

The mesh equations. 80

$$I_2 - I_1 = 2$$

$$-80 + 6I_1 + 10I_2 + 4I_x = 0$$

$$I_x = I_1$$

$$I_2 - I_1 = 2 \text{ --- ①} \quad \text{بحوضهم على الآلة حاسبة}$$

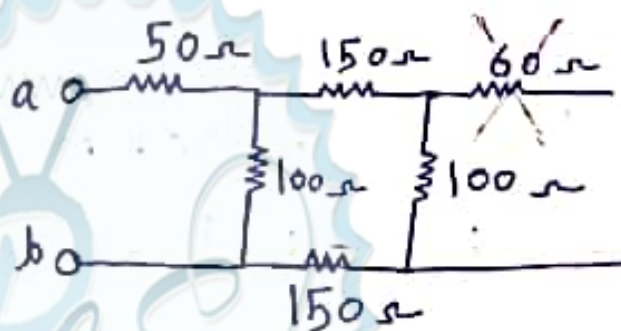
$$10I_2 + 10I_1 = 80 \text{ --- ②}$$

$$I_2 = 5A \text{ [2]} \quad I_1 = I_x = 3A \text{ [1]}$$

$$\text{[3]} \quad p = VI = 80 \times 3 = 240$$

4 * يحل بطريقتين:

- ١- على توصيلة Δ و Y
- ٢- بلغي المدة (لأن الدارة مفتوحة لا يمر فيها تيار) و يحل توالي وتوازي.



$$150, 100, 150 \xrightarrow{\text{توالي}} R_1 = 400 \text{ } \Omega$$

$$R_1, 100 \xrightarrow{\text{توازي}} R_2 = 80 \text{ } \Omega$$

$$R_2, 50 \xrightarrow{\text{توالي}} R_{eq} = 130 \text{ } \Omega$$

Don't miss the change ... Be part of it

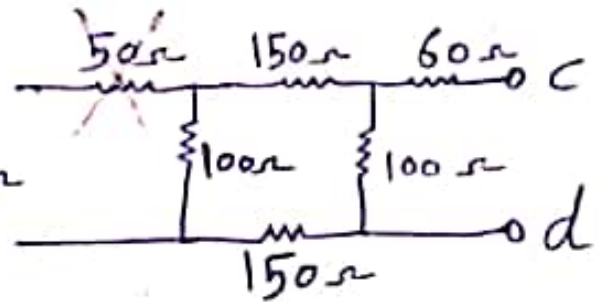


أسئلة سنوات سابقة

المادة: سيركت 1

[5]

نفس طريقة [4]



$$150, 100, 150 \xrightarrow{\text{توالي}} R_1 = 400 \Omega$$

$$R_1, 100 \xrightarrow{\text{توازي}} R_2 = 80 \Omega$$

$$R_2, 60 \xrightarrow{\text{توالي}} R_{eq} = 140 \Omega$$

[6] $I = 30 \text{ A}, V = 200 \text{ V}, t = 5 \text{ s}$

$$W = P \times t = VI \times t$$

$$= 200 \times 30 \times 5 = 30 \text{ KJ}$$

[7] $q = (10 - 10e^{-2t}) \text{ mC}, t = 1 \text{ s}$

$$I = \frac{dq}{dt} = 20e^{-2t}$$

$$I(1) = 20e^{-2} = 2.707 \text{ mA}$$

[8] $G \rightarrow \text{Giga} = 10^9$

PCE-TEAM

Don't miss the change ... Be part of it



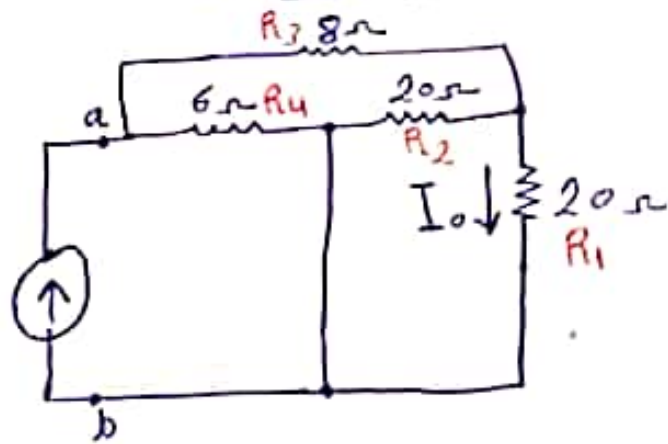
أسئلة سنوات سابقة

المادة: سيركيت 1

9 $R_1, R_2 \xrightarrow{\text{توازي}} R_5 = 10 \Omega$
 ← نفس البداية
 ونفس النهاية

$R_5, R_3 \xrightarrow{\text{توالي}} R_6 = 18 \Omega \quad 16A$

$R_6, R_4 \xrightarrow{\text{توازي}} R_{eq} = 4.5 \Omega$



10 $R_{eq} = 4.5 \Omega, I_{eq} = 16A$

$V_{eq} = I \times R = 72V = V_4 = V_6 \Rightarrow$ توازي

$I_6 = \frac{V_6}{R_6} = 4A = I_3 = I_5 \Rightarrow$ توازي

$V_5 = I_5 \times R_5 = 40V = V_2 = V_1 \Rightarrow$ توازي

$I_1 = \frac{V_1}{R_1} = 2A = I_0$

11 $V_{8\Omega} = V_3 = I_3 \times R_3 = 32V$

The nodal equations:

$\frac{V_1 - V_2}{6} + \frac{V_1}{2} = 2 \dots \textcircled{1}$

$i_0 = \frac{V_1 - V_2}{6} \dots \textcircled{2}$

$\frac{V_2 - V_1}{6} + 3i_0 + 2 + \frac{V_2 - 24}{8} = 0 \dots \textcircled{3}$

Don't miss the change ... Be part of it



أسئلة سنوات سابقة

المادة: سيركت ١

$$\frac{2}{3} V_1 - \frac{1}{6} V_2 = 2 \quad \text{--- (1)}$$

$$\frac{V_2 - V_1}{6} + 3 \left(\frac{V_1 - V_2}{24} \right) + \frac{V_2 - 24}{8} + 2 = 0$$

$$\frac{1}{3} V_1 - \frac{5}{24} V_2 = 1 \quad \text{--- (2)}$$

$$V_1 = 3V \quad , \quad V_2 = 0V$$

12 13

$$\boxed{14} \quad i_o = \frac{V_1 - V_2}{6} = 0.5A$$

$$\boxed{15} \quad V_{2\Omega} = V_1 = 3V$$

$$P = \frac{V^2}{R} = 4.5W$$

$$\boxed{16} \quad I_{24V} = \frac{V_2 - 24}{8} = -3A$$

$$P = V * I = 24 * 3 = 72W$$

PCE-TEAM

Don't miss the change ... Be part of it